Exploration of Tridentate Ligands on Halogen-terminated Hybrid MXenes

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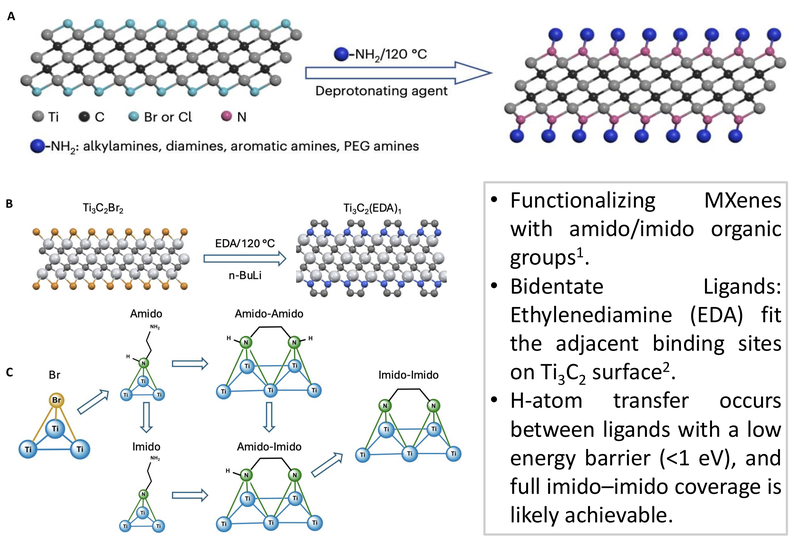
Monodentate and Bidentate Ligands



- Functionalizing MXenes with amido/imido organic groups[1].
- Bidentate Ligands: Ethylenediamine (EDA) fit the adjacent binding sites on Ti_3C_2 surface[2].
- H-atom transfer occurs between ligands with a low energy barrier (<1 eV), and full imido–imido coverage is likely achievable.

Fig 1. (A) The Synthesis Scheme of monodenate liganded hybrid MXene (h-MXene).
(B) The functionalization of amido- & imido- group in bidentate ligands
(C) The Synthesis Scheme of bidenate liganded h-MXene.

Project Purpose

This project highlights the first exploration of **tridentate ligands bonding** onto **MXene surfaces**. Using **DFT** and **AIMD** simulations, we evaluate their binding strength, reaction behavior and molecular dynamics.

Acknowledgement

This research is supported by the NSF Center for Chemical Innovation – MXenes Synthesis, Tunability and Reactivity (M-STAR). Computational resources at Vanderbilt University were used for calculations.

M-STAR

NSF Center for MXenes Synthesis, Tunability, and Reactivity

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References

[1]Zhou, C., Wang, D., Lagunas, F. *et al.* Hybrid organic–inorganic two-dimensional metal carbide MXenes with amido- and imido-terminated surfaces. **Nat. Chem. 15**, 1722–1729 (2023). https://doi.org/10.1038/s41557-023-01288-w

[2]Zhou, C.; Kim, Y-H.; Atterberry, B.; Khokhar, V. et al. *MXenoids: Generalization of MXene-Inspired Covalent Surface Modifications Across Two-Dimensional Materials.* **J. Am. Chem. Soc. 2025**, *147* (27), 23743–23757. https://doi.org/10.1021/jacs.5c05771

Design and Binding of Triamine Ligands

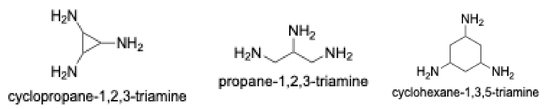


Figure 2. Structures of the 3 triamine studied for MXene triligand binding, featuring symbolic triangular, linear, cyclic carbon backbones.

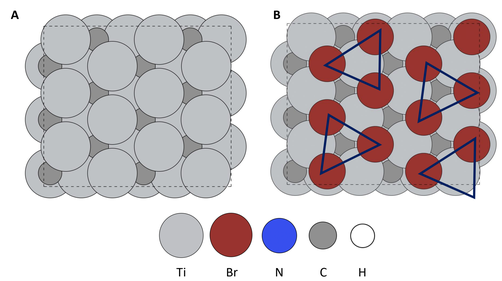


Figure 3. Structures of the (A) Ti_3C_2 MXene surface and (B) $Ti_3C_2Br_2$ MXene where blue-highlighted triangles: formation site of tridentate ligands on fcc hollow sites.

Binding Energies: Bond Strength

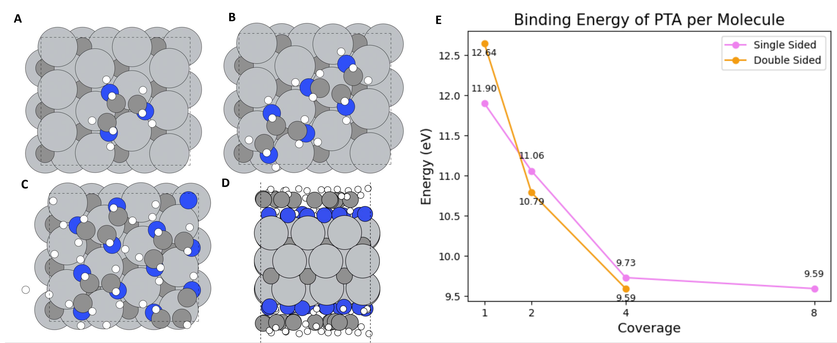


Fig 4. The optimized structure of PTA on the termination onto bare MXene Ti_3C_2, showing (A) Coverage 1; (B) Coverage 2; (C) Coverage 4; (D) Coverage 8;
(E) The Binding Energy of PTA Tridentate ligand /molecule with increasing coverages. Single-sided models were used for comparison, while double-sided systems better represent realistic surface environments.

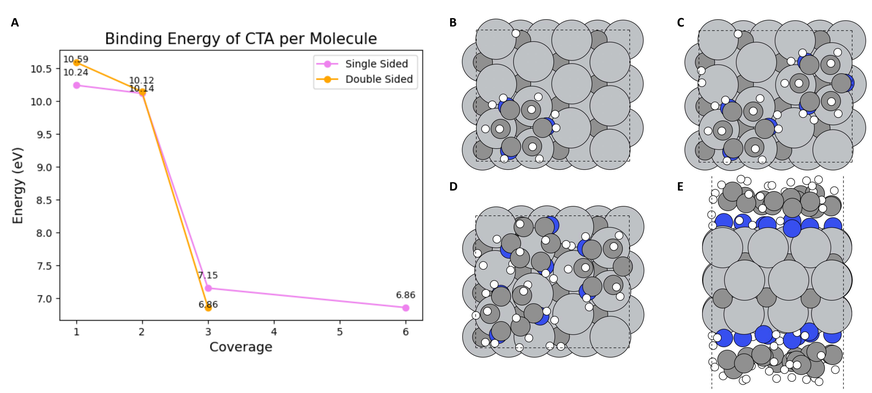


Fig 5. The optimized structure of CTA on the termination onto bare MXene Ti_3C_2, showing Nitrogen (blue) binding onto fcc hollow sites
(A) The Binding Energy of CTA Tridentate ligand per molecule with increasing coverages; (B) Coverage 1; (C) Coverage 2; (D) Coverage 3; (E) Coverage 6;

Reaction Energies: Reactivity

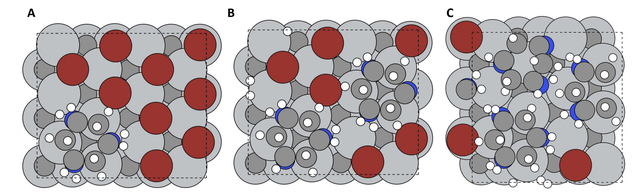


Figure 6. Optimized Structure of CTA binding on MXene with 1,2,3 coverages

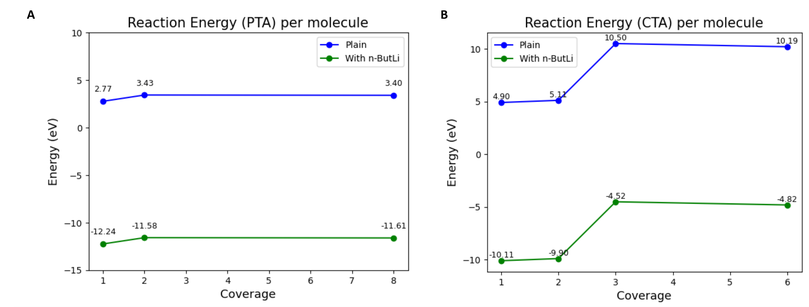


Figure 7. Reaction energy per molecule for PTA (A) and CTA (B) ligands bound single-sided onto $Ti_3C_2Br_2$ surfaces via nucleophilic substitution, with and without the strong deprotonating reagent *n*-butyllithium (n-ButLi).
Blue represents direct substitution, green shows significant thermodynamic spontaneity under n-ButLi, indicating enhanced need of strong base in deprotonating stage. .

Molecule Dynamics of Tridentate liganded MXenes

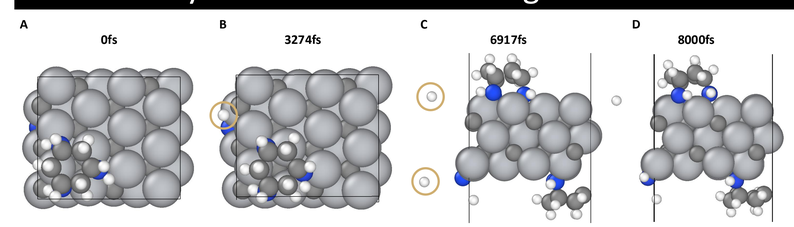
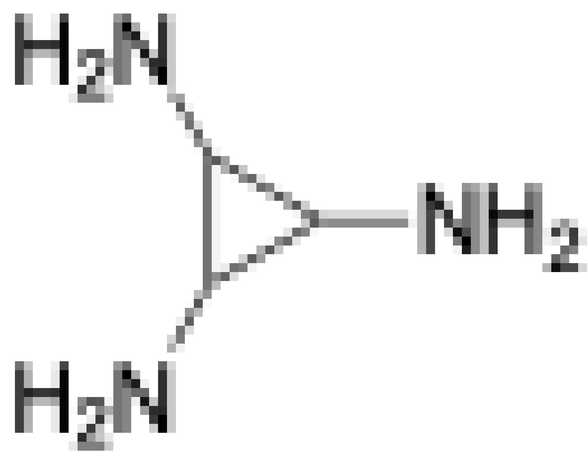


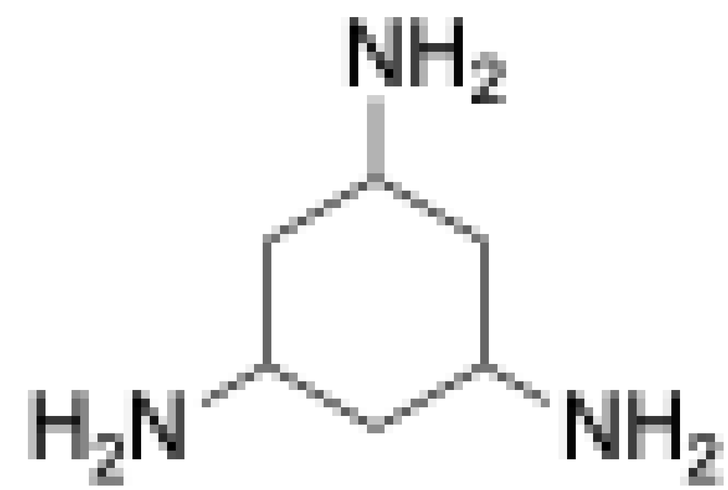
Figure 8. Snapshots of AIMD simulation at 400 K of CTA tridentate monolayer with coverage 1 on Ti_3C_2 at (A) 0 fs, (B) 3274 fs; circle highlighting H transfer and turning –NH into –N, and surface hydride (C) 6917 fs; circle featuring double-sided H has been transfer from original group , (D) 8000 fs. By the end of timeframe, Hydrogen from two sides separated from original –NH groups forming amido to imido terminations and surface hydrides on fcc sites.

Conclusion

- **Tridentate ligands** succeeded in terminating MXenes, occupying closest triangular fcc sites.
- **Binding energies** of Tridentate ligands remain **consistent** with the **bidentate** ligands[2] , indicating similar stable coordination to MXene surface.
- The reaction **requires** strong **deprotonating agents** (e.g., *n*-butyllithium) to proceed thermodynamically spontaneous.
- AIMD simulations **confirmed proton transfer** (–NH to –N and –H) during nucleophilic substitution, supporting the proposed reaction mechanism.

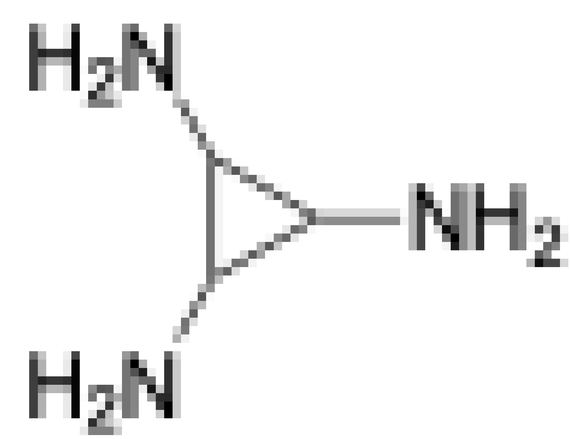


cyclopropane-1,2,3-triamine



cyclohexane-1,3,5-triamine

propane-1,2,3-triamine



cyclopropane-1,2,3-triamine